

Abhinav Pratap Singh

4th year PhD, Georgia Tech

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Research Interest

A fourth year PhD student interested in imaging the earth primarily via electromagnetic methods. This can lead to applications in investigating properties of mantle and deeper earth, carbon capture (CCS), and exploration of geothermal reservoirs. To bring these to effect, my research is inclined towards (but not limited to) geophysical inverse problems, scientific machine learning, uncertainty quantification and mixing information from different types of datasets..

Education

Georgia Institute of Technology

PhD in Earth and Atmospheric Sciences, CGPA: 4.0/ 4.0

Atlanta, GA

Aug 2022 - Present

Indian Institute of Technology (Indian School of Mines) Dhanbad

Integrated MTech in Applied Geophysics, CGPA: 8.93/ 10.0

Dhanbad, India

July 2017 - May 2022

Modern Academy Inter College

12th grade (US equivalent: High School), 98% (All India Rank: 7)

Lucknow, India

July 2007 - June 2017

Invited Talks

Accelerating Uncertainty Quantification of Magnetotelluric (MT) data using Neural Operators

Abhinav Pratap Singh, Samer Naif

Society of Industrial and Applied Mathematics (SIAM) : Uncertainty Quantification, 2026

Constraining the upper mantle properties in the Pacific seafloor using MT and seismic data

Abhinav Pratap Singh, Samer Naif, Samuel Ofori, Joshua B Russel

Gordon Research Seminar : Interiors of the Earth, 2025

Conference Proceedings

Imaging the conductivity structure of the Kivu rift system using stochastic inversion of magnetotelluric data

Abhinav Pratap Singh, Samer Naif, Andrew Vern Newmann, Derrick Murekezi, Tite Niyetegeka, Cynthia J Ebinger

AGU Fall Meeting Abstract, 2024

Magnetotelluric imaging of the Kivu rift system

Abhinav Pratap Singh, Samer Naif, Tite Niyetegeka, Andrew Vern Newmann, Derrick Murekezi, Cynthia J Ebinger

Electromagnetic Induction Workshop, 2024

Integrating seismic and MT data to constrain upper mantle properties beneath the Pacific plate

Abhinav Pratap Singh, Samuel Ofori, Samer Naif, Joshua B Russel

AGU Fall Meeting Abstract, 2024

Earthquake Analysis with Virtual Seismograms: Leveraging Symmetric Autoencoders for Time-Reversal Imaging

Isha Lohan, Madhusudan Sharma Sharma, Abhinav Pratap Singh, Pawan Bharadwaj

AGU Fall Meeting Abstract, 2023

Preliminary results from a magnetotelluric survey of Kivu rift system

Abhinav Pratap Singh, Samer Naif

AGU Fall Meeting Abstract, 2023

A deep learning approach for joint inversion of DC Resistivity and MT data

A.P. Singh, D. Vashisth, S. Srivastava

82nd EAGE Annual Conference Exhibition, 2021

Petrophysical Inversion of seismic dataset using artificial neural networks

Abhinav Singh, Divakar Vashisth, Anmol Vishwakarma

AGU Fall Meeting Abstracts, 2021

Deep learning for joint geophysical inversion of seismic and MT data sets

Abhinav Pratap Singh, Divakar Vashisth, Shalivahan Srivastava

First International Meeting for Applied Geoscience Energy Expanded Abstracts, 2021

One-Dimensional Seismic Inversion using Machine Learning techniques

P. Singh, D. Vashisth, A. P. Singh

AGU Fall Meeting Abstracts, 2020

Random Forest Regressor for Layered Earth Data Inversion

A. P. Singh, D. Vashisth, S. Srivastava

Research Experience and Other Projects

Porosity.jl, a julia package for differentiable rock physics modeling and inference :

June 24 - Present

Appliaction to LAB system at NoMelt

Georgia Institute of Technology

I am developing Porosity.jl, an open-source Julia package to perform Bayesian inference of geophysical observables (seismic velocities and electrical conductivities) to obtain the rheological parameters of the system. The package is being subsequently used to quantify the rheological properties at mid-ocean Pacific seafloor and learn more about the Lithosphere-Asthenosphere Boundary by integrating thermodynamics into the study.

Imaging the electrical resistivity structure of the Kivu rift system

May 23 - Present

Georgia Institute of Technology

To image the magmatic plumbing Kivu rift system, we scouted and deployed the MT stations in Rwanda. The hilly terrain, high population densities, and power lines in the valleys made this acquisition challenging. Consequently, the acquired data set required rigorous processing. I developed the subsurface conductivity structure and the associated uncertainties using Bayesian inference. The obtained uncertainties were then propagated to estimate rock physics properties of the conductors.

PrISM.jl, a julia package for differentiable rock physics modeling and inference :

Jan 23 - Present

Appliaction to LAB system at NoMelt

Georgia Institute of Technology

I am developing PrISM.jl, an open-source Julia package to model different types of geophysical datasets. The package provides support to perform deterministic and probabilistic inversion using automatic differentiation. The package has been used in all my research to perform inversions, both in the Kivu-rift system as well as the NoMelt dataset.

Developing NeuralOperators.jl, a julia package to provide neural operators

May 24 - July 24

Google Summer of Code

Contributed to developing NeuralOperators.jl, a julia package to provide high-performance codes for neural operators. I contributed towards the structure of the package, implemented DeepONet and NoMAD (Nonlinear Manifold Decoder) from scratch, and made the documentation. We have already obtained a $\sim 10x$ speedup against the python package for DeepONets.

A scalable and high performance algorithm to obtain earthquake source images using backprojection

Jan 22- July 22

Indian Institute of Science, Bangalore

Developed a high performance code, scalable to GPUs, to use the rupture propagation images of the earthquakes using the backprojection strategy, modified later into Time reversal algorithm. The codes will be used on virtual seismic traces to get images of the earthquakes after de-Dopplerization of the waveforms.

Potential investment opportunity and carbon mitigation measures in South Pelto Area, Gulf of Mexico

Jan 21- Sept 21

SEG EVOLVE' 21

Team Lead of the IIT (ISM) Dhanbad team, combined with the IIT Roorkee and the Carbon Capture Coordinator of the combined team.

Contributed to seismic data interpretation and consequently located reservoirs for investment and CO2 storage to find the best investment opportunity in the South Pelto Area, Gulf of Mexico.

Field Experiences

Land MT data acquisition on Kivu rift

Summers 2023

Rwanda

I was involved in land MT data acquisition in Rwanda to understand the magmatic plumbing in the rifting regime. I assisted in deploying, collecting and scouting for sites during a 3-week long project. The hilly terrain, high population densities, and power lines in the valleys made this acquisition challenging

Marine MT data acquisition over Cocos plate

Fall 2025

Pacific Ocean, offshore Costa Rica

I was part of the crew that deployed Ocean Bottom MT instruments over the Pacific seafloor to image the extent of melt channel in the subsurface. I was engaged in setting up, deployment and recovery of the MT instruments over the period of 4 weeks.

Teaching Experiences

Graduate Teaching Assistant, Georgia Tech

Spring 2023

EAS 2655: Quantitative Techniques

- Gave 2 short lectures on Linear Algebra and introduction to programming in Python and Matlab
- Graded the assignments and provided detailed feedback and comments

Graduate Teaching Assistant, Georgia Tech

Fall 2023

EAS 6130: Earth Systems Modeling

- TAed one of the most challenging grad level courses in EAS, GaTech
- Graded the assignments and provided detailed feedback and comments
- Organized one-on-one discussions with students
- Received outstanding feedback on my TAing abilities

Awards and Achievements

- Awardee of **Society of Exploration Geophysicists (SEG) Foundation, SEG/Frans and Alice Hammons Scholarship** for the academic year 2025-26.
- Awardee of **Society of Exploration Geophysicists (SEG) Foundation, SEG/ Gerald W. Hohmann Memorial, and SEG/Norman and Shirley Domenico Scholarship** for the academic year 2024-25.
- Awardee of **SEG Foundation/Apache scholarship**, given by Society of Exploration Geophysicists (SEG) for the academic year 2023-24.
- Awardee of **SEG Foundation/Chevron scholarship**, given by Society of Exploration Geophysicists (SEG) for the academic year 2021-22.
- Awardee of **Indian School of Mines Alumni Association North America chapter (ISMAANA) award**, given to top 2 students from the Department of Applied Geophysics, for the academic year 2020-21.
- Awardee of **L Austin Weeks Undergraduate Grant**, granted by American Association of Petroleum Geologists (AAPG)
- Awardee of **Merit-cum-Means Scholarship**, from the Govt. of India consecutively from 2018 to 2021.
- In Switch Energy International Case Competition 2020, organised by Switch Energy Alliance, my team "Lalten" made it to the Finals, and secured the 6th position in the event wherein more than 250 teams registered from all the inhabited continents of the world.
- One of the 53 students selected around the world for the SEG Student Leadership Program (SLS), 2021 held in online mode.

Skills

• Technical Skills

- Extensive background in programming in languages like Julia, Python, Matlab, specializing in Scientific computing, Data Visualisation, Data Processing, Machine Learning, and Neural Networks.
- Well versed in seismic data processing and interpretation softwares like Petrel and OpendTect.

- **Soft Skills** Having managed two clubs, multiple teams and organised numerous events in the college, I have a good experience in teamwork, leadership, and public speaking.

- **Hobbies** Reading and writing (maintaining an blog). A current member of Book Jackets, Georgia Tech, and a past member of The Book Club and LitC, The Literary and Debating Club of IIT (ISM) Dhanbad.

• Extra-curriculars

- Gold Medalist in District Boxing Championship and Inter Hostel Boxing Championship in the lightweight category.
- I was also part of the Tare Zameen Foundation, a non-profit organisation that works for the uplifting of children, women, aged and mentally challenged with a focus on Health, Education, and Empowerment.

Positions of Responsibility

May'24 - May'25	Vice President at GEAS (Graduate Student in Earth & Atmospheric Sciences), Georgia Tech)
Aug '23- May'24	Social Chair at GEAS (Graduate Student in Earth & Atmospheric Sciences), Georgia Tech)
March '20- March '21	President at SEG IIT (ISM) Student Chapter
March' 19- March '20	Vice President at EAGE IIT (ISM) Student Chapter
Aug' 19- Aug '20	Coordinator of Book Club, IIT (ISM) Dhanbad
March '18- March '19	Event Manager at SEG IIT(ISM) Student Chapter